JI YEON KIM

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Research Portfolio: https://jiyeon-kim-ko.github.io/porfolio/

PROFESSIONAL SUMMARY

Pharmaceutical formulation researcher with an M.Eng. in Industrial Pharmaceutics and experience in drug product development, process optimization, and pharmaceutical quality assessment. Applied Quality by Design (QbD), Design of Experiments (DoE), Process Analytical Technology (PAT), multivariate analysis, and AI-based modeling to investigate relationships between manufacturing variables and critical quality attributes (CQAs).

RESEARCH HIGHLIGHTS 10 SCI(E) papers (9 first/co-first) | 10 national R&D projects | 5+ CTD 3.2.P dossiers

Areas of Expertise

- Formulation Development
- Quality by Design (QbD)
- Process Analytical Technology (PAT)
- Drug Delivery Systems
- Process Development & Optimization
- Design of Experiments (DoE)
- Multivariate Data Analysis
- Quality Prediction & AI Modeling

EDUCATION

Daegu Catholic University | Gyeongsan, South Korea Mar 2023 – Feb 2025
Doctoral Studies in Pharmacy with GPA 4.5 / 4.5 — coursework and qualifying examination completed; Ph.D. degree not awarded.

Inje University | Gimhae, South Korea Aug 2021 – Feb 2023
Master of Engineering (M.Eng.) with GPA 4.5 / 4.5, Industrial Pharmaceutics

Inje University | Gimhae, South Korea Mar 2018 – Aug 2021
Bachelor of Engineering (B.Eng.) with GPA 4.07 / 4.5, Pharmaceutical Engineering

PROFESSIONAL EXPERIENCE

KI Bio Co., Ltd. | Formulation R&D Researcher Mar 2025 – Dec 2025

Transdermal Patch Development

- Developed drug-in-adhesive patches for Parkinson's disease treatment by evaluating polymer and permeation-enhancer combinations to optimize drug release and skin permeation.
- Redesigned the patch preparation process to address drug content variability and assessed formulation performance through drug release, skin permeation, and accelerated stability testing.

QbD-Based Valsartan Tablet Development

- Optimized valsartan tablet formulation and manufacturing parameters using a Quality by Design (QbD) approach to improve formulation and process performance.
- Validated an HPLC analytical method and evaluated dissolution profiles against the reference product to confirm dissolution equivalence.

PAT & AI-Enabled Fixed-Dose Combination Development

- Developed a triple fixed-dose combination bilayer tablet and established a process control strategy using Process Analytical Technology (PAT), including NIR, Eyecon, torque measurements, and digital image correlation (DIC).
- Developed an artificial neural network (ANN) model linking in-process measurements to finished-product quality attributes ($R^2 > 0.9$).

RELEVANT FORMULATION & PROCESS RESEARCH PROJECTS

Mirabegron Extended-Release Formulation 2022–2023

- Optimized extended-release tablet formulations using D-optimal mixture design to investigate the effects of formulation variables on drug release.
- Investigated swelling and erosion mechanisms and supported granulation scale-up from laboratory to pilot scale.

LED-PIV Powder Flow Analysis 2025–2026

- Developed an LED-based particle image velocimetry (PIV) method to characterize dynamic powder flow behavior across six pharmaceutical excipients.
- Compared PIV-derived measurements with FT4 powder rheometer to evaluate powder flow characteristics.

ADDITIONAL RESEARCH PROJECTS

Pilot-Scale Wet Granulation and Process Analysis

- Evaluated impeller speed, liquid-to-solid ratio, and spray time during pilot-scale mirabegron granulation; monitored blend content by NIR and sampled particle size.

Ocular Drug-Delivery & Formulation Research

- Investigated contact lens-based drug-delivery systems, including formulation design, physicochemical assessment, and drug-release evaluation.
- Evaluated drug release and ocular permeation, and developed an accessory to improve contact-lens release testing.

SELECTED PUBLICATIONS

Integrated control strategy based on deep learning and process analytical technology for the manufacture of a triple fixed-dose combination tablet

Journal of Pharmaceutical Investigation | *2025*

Developed an integrated control strategy combining Quality by Design (QbD), Process Analytical Technology (PAT), and deep learning to optimize the manufacturing processes of a triple fixed-dose combination tablet.

Key Methods

- Quality by Design (QbD)/Design of Experiments (DoE)
- Process Analytical Technology (PAT)
- High-Shear Wet Granulation and Roller Compaction
- Artificial Neural Network (ANN)

Developmental strategy for swellable/erodible matrix tablet of mirabegron: quality by design approach with various geometric properties and pharmacokinetic evaluation

Journal of Pharmaceutical Investigation | *2023*

Developed and optimized a sustained-release matrix tablet formulation of mirabegron using a QbD approach. Investigated the relationships between formulation variables, swelling and erosion behavior, and drug release, and evaluated the optimized formulation through in vitro dissolution and in vivo pharmacokinetic studies.

Key Methods

- Quality by Design (QbD
- Swelling, Erosion Kinetics, and Gel Strength Analysis
- Formulation Optimization
- In Vitro Dissolution/In Vivo Pharmacokinetic Evaluation

Deep learning-based image classification and quantification models for tablet sticking

International Journal of Pharmaceutics | *2025*

Developed an integrated image analysis model combining CNN-based classification and GLCM–SVM-based quantification to detect and assess tablet sticking during pharmaceutical manufacturing. Achieved 99.39% classification accuracy using GoogLeNet.

Key Methods

- Convolutional Neural Networks (CNNs)
- Support Vector Machine (SVM)
- Gray-Level Co-occurrence Matrix (GLCM)
- Sticking Index Quantification

Selected publications; complete bibliography available on request.

REGULATORY & SCIENTIFIC WRITING

- Drafted CTD Module 3.2.P documentation for five or more projects and developed analytical methods and national R&D reports.
- Prepared scientific manuscripts and technical reports documenting formulation development, PAT, and quality-prediction research.

SKILLS & ADDITIONAL INFORMATION

- Analytical & PAT: HPLC, UV–Vis, NIR, FT-IR, Eyecon, Digital Image Correlation (DIC), Particle Image Velocimetry (PIV), FT4 powder rheometer.
- Formulation & Process Techniques: Granulation, tablet compression, coating, Franz diffusion cell testing, dissolution testing, stability studies.
- Statistical & Computational Software: Design-Expert, MODDE, SIMCA, JMP, Minitab, MATLAB, Python, R, SPSS, Origin.
- Languages**:** Korean (Native), English (TOEIC Speaking AL)